AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

[Sales and Business Data Cleaning & Visualization Using Excel]

[SUNLYTICS 2024–2026]

Project Domain

[Microsoft Excel & Power BI with Copilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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I am also grateful to Rishi UBR Women's College for the infrastructure and support provided during the project. Finally, I thank my family and peers for their constant encouragement.

ABSTRACT OF THE PROJECT

This project addresses the challenge of unstructured and inconsistent solar panel sales data, which prevents businesses from making informed decisions. The dataset (2024–2026) contained missing values, inconsistent formats, and duplicate records. Using Microsoft Excel, Power BI, and Copilot, we cleaned and standardized the dataset, applied pivot tables and formulas for analysis, and built an interactive dashboard.

Key insights include:

- Seasonal peaks in solar panel sales.
- Efficiency advantages of monocrystalline panels vs. cost benefits of polycrystalline panels.
- Differences between residential and commercial market segments.

The outcomes support inventory optimization, targeted marketing, and product strategy decisions. This project demonstrates how AI-assisted Excel and Power BI workflows can transform raw data into actionable business intelligence.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

Solar panel distributors collect large amounts of sales and performance data, but it is often unstructured, inconsistent, and difficult to analyse. Without proper cleaning and visualization, businesses cannot identify sales trends, panel performance, or market segmentation, leading to poor inventory planning and missed revenue opportunities

1.2 Motivation

The solar energy market is rapidly expanding, and structured insights are essential for competitiveness. This project is meaningful because it equips businesses with clarity on sales trends and helps students gain practical skills in AI-driven analytics.

1.3 Objectives

Clearly state the objectives of your project. Objectives should be measurable and specific.

1. To collect and understand the structure of the dataset used in the project.
2. To clean and preprocess raw data by removing duplicates, correcting formats, and handling missing values.
3. To apply domain-specific analysis using Excel, Power BI with Copilot.
4. To build interactive dashboards or reports that clearly communicate insights.
5. To leverage AI tools Copilot to automate or enhance the analysis and content generation process.
6. To Clean and standardize raw solar panel sales data.
7. To Visualize sales trends over time.
8. To Compare monocrystalline vs. polycrystalline performance.
9. To Analyse residential vs. commercial segments.
10. Build an interactive dashboard for decision-makers.

1.4 Scope of the Project

Define the boundaries and limitations of the project clearly.

- This project covers data from solar panel sales records collected between 2024 and 2026, including sales transactions, panel types, customer segments, and performance metrics.
- The tools used are limited to Microsoft Excel, Power BI, and Copilot AI as assigned for data cleaning, visualization, and AI-assisted analysis.

- The analysis focuses on data cleaning, visualization, report generation, and prompt engineering to extract actionable business insights.
- The project does not include real-time data integration, predictive machine learning models, or deployment as a production-level system.
- This project is intended as a fellowship showcase to demonstrate practical skills in AI-driven data analytics and is not designed for operational use.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

Several studies and industry practices have explored the importance of structured data cleaning and visualization in business intelligence. Microsoft Excel has long been used for handling inconsistencies in datasets through functions such as IFERROR, COUNTIF, and pivot tables. Power BI templates have been widely adopted to create interactive dashboards that highlight revenue trends, product performance, and regional segmentation.

In addition, recent research on AI-assisted data cleaning emphasizes the role of intelligent tools like Microsoft Copilot in automating repetitive tasks, generating summaries, and identifying anomalies. Studies on prompt engineering frameworks show that optimized AI prompts can significantly improve the quality of insights generated by tools such as Copilot or ChatGPT. Business intelligence dashboards, as documented in IEEE publications, have proven effective in transforming raw data into actionable insights for decision-makers.

2.2 Tools and Technologies Review

Tools Used

- **Microsoft Excel**: Utilized for data cleaning, transformation, and preliminary analysis using features like pivot tables, conditional formatting, and data validation.
- **Microsoft Power BI**: Used to create interactive dashboards and visualizations for effective business intelligence and performance analysis.
- **Microsoft Copilot (AI)**: Assisted in automating tasks, generating summaries, and supporting data analysis through natural language inputs.
- **AI Chatbots (e.g., ChatGPT, Gemini)**: Applied for content generation, report drafting, and prompt-based analytical support.

Models, Techniques, and Approaches

- **Excel-Based Data Cleaning Methods**: Applied for preprocessing, removing duplicates, and ensuring data consistency.
- **Power BI Dashboarding Techniques**: Used to design interactive and insightful visual reports.
- **AI Prompt Engineering**: Implemented to improve the accuracy and relevance of AI-generated outputs.
- **Business Intelligence Practices**: Adopted to present complex data in a clear, structured, and user-friendly manner.

2.3 Gaps Identified

- Existing research mainly focuses on **advanced machine learning techniques**, with limited attention to **simple, tool-based solutions** suitable for non-technical users.

- There is a lack of **practical integration studies** involving Microsoft Copilot with Excel and Power BI for domain-specific analysis.
- **Student-level project implementations** using these tools are not widely documented.
- **Standardized frameworks for prompt engineering** in business intelligence are still in the early stages of development.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

- **Data Source / Input**: Raw dataset in CSV format containing structured and unstructured data relevant to the project analysis.
- **Data Cleaning**: Performed using Microsoft Excel by removing duplicates, handling missing values, correcting data types, and standardizing formats to ensure data consistency and accuracy.
- **Analysis / Processing**: Conducted using pivot tables, Excel functions, and Microsoft Copilot prompts to identify patterns, trends, and key insights.
- **Visualization / Output**: Developed an interactive dashboard in Microsoft Power BI to present insights, trends, and performance metrics in a clear and visually engaging manner.

3.2 Modules / Components Used

List the key modules or functional components of your project:

- **Module 1 — Data Ingestion**
  The raw solar panel sales dataset (2024–2026) was imported into Microsoft Excel and Power BI. This step ensured that all records, including attributes such as panel type, wattage, efficiency, voltage, and current, were available for processing.

- **Module 2 — Data Cleaning**
  Inconsistencies, duplicates, and missing values were identified and corrected. Excel formulas (`IFERROR`, `COUNTIF`), data validation, and Copilot assistance were used to standardize formats and prepare the dataset for analysis.

- **Module 3 — Data Analysis**
  Pivot tables, charts, and formulas were applied to explore sales trends, compare monocrystalline vs. polycrystalline performance, and analyze residential vs. commercial segments. Copilot prompts were used to generate summaries and highlight anomalies.

- **Module 4 — Visualization / Report Generation**
  Interactive dashboards were built in Power BI and Excel to present insights in a clear, business-friendly format. Visuals included monthly sales trends, efficiency comparisons, and segmentation charts.

- **Module 5 — AI Integration**
  Microsoft Copilot was integrated to automate repetitive tasks, generate professional summaries, and assist in identifying seasonal peaks and product performance patterns. This reduced manual effort and improved accuracy.

- **Module 6 — Business Outcomes (Project-Specific)**
  Insights from the dashboards were translated into actionable strategies for inventory optimization, targeted marketing, and product planning. This module ensured that the technical analysis directly supported business decision-making.

- ## 3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) illustrates how data moves through the system, from raw input to final business insights. It helps visualize the processes, transformations, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

At the highest level, the system is represented as a single process. The raw dataset is the input, and the cleaned dataset, dashboards, and reports are the outputs.

Flow Chart:

This diagram shows the overall transformation of unstructured data into actionable intelligence.

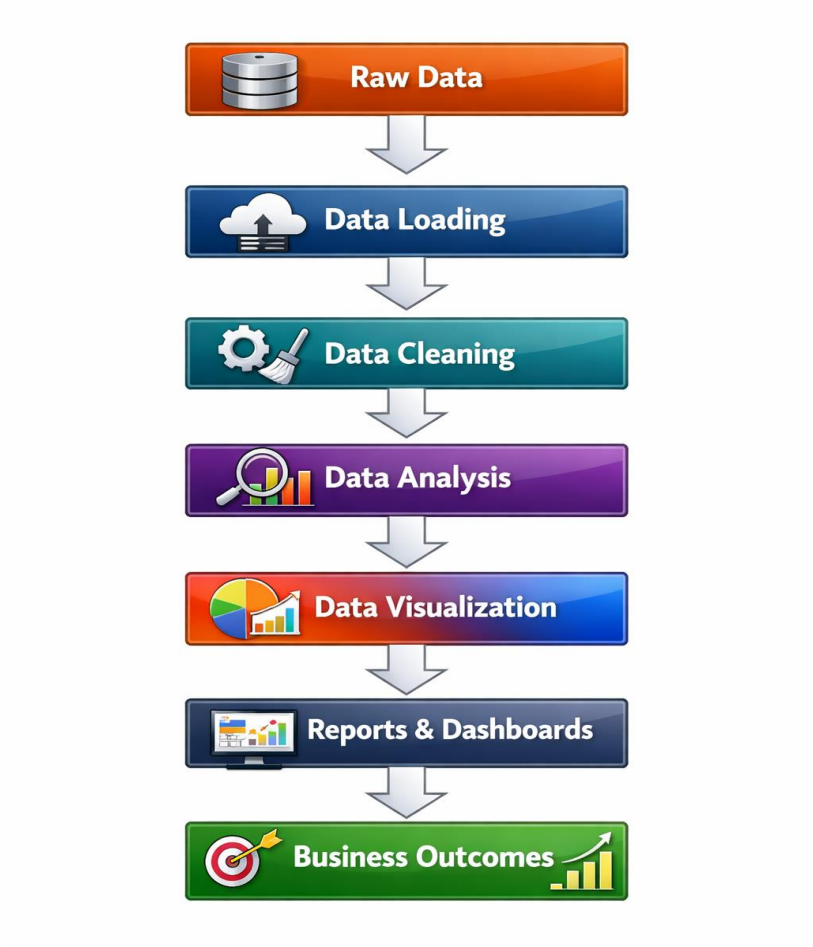
DFD Level 1 — Detailed Process Flow

This level breaks down the main process into sub-processes, showing how data is handled step by step:

1. **Data Loading and Validation**
    - Import raw dataset into Excel / Power BI.
    - Validate field formats (dates, wattage, efficiency).
2. **Data Cleaning and Transformation**
    - Remove duplicates and null values.
    - Standardize formats using Excel formulas and Copilot assistance.
    - Ensure consistency across panel types and segments.
3. **Analysis and AI Processing**
    - Apply pivot tables, charts, and formulas to identify sales trends.
    - Use Copilot prompts to summarize seasonal peaks and anomalies.
    - Compare monocrystalline vs. polycrystalline performance.
4. **Visualization and Report Generation**

- Build interactive dashboards in Power BI and Excel.
- Generate charts for residential vs. commercial segmentation.
- Produce executive-friendly reports summarizing insights.

Flow:



DFD Level 0 — Context Diagram

Shows the system as a single process with its inputs and outputs.

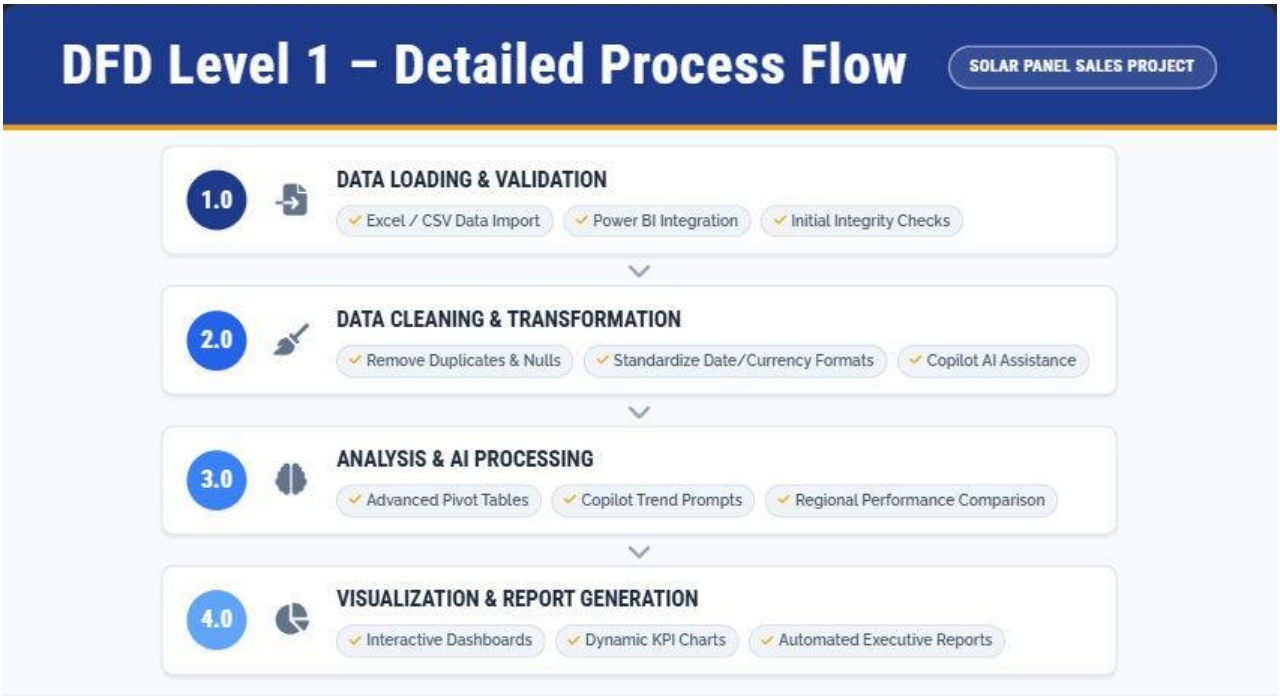
[Insert your DFD Level 0 diagram here. The diagram should show: Raw Data Input → [Your Project System] → Cleaned Data / Dashboard / Report Output]

DFD Level 1 — Detailed Process Flow

Breaks down the main process into sub-processes. For your project this may include:

- Sub-process 1: Data Loading and Validation
- Sub-process 2: Data Cleaning and Transformation
- Sub-process 3: Analysis and AI Processing

- Sub-process 4: Visualization and Report Generation



3.4 Advantages

The proposed methodology for cleaning and visualizing solar panel sales data offers several distinct advantages:

- **Automated Data Preparation**
  By using Excel formulas and Copilot assistance, repetitive tasks such as handling missing values, correcting formats, and removing duplicates are automated. This reduces manual effort and ensures consistency across the dataset.

- **Clarity in Business Insights**
  Interactive dashboards in Power BI present complex sales and performance data in a simple, visual format. Managers and non-technical stakeholders can easily interpret seasonal trends, panel efficiency comparisons, and market segmentation.

- **Improved Decision-Making**
  The analysis directly supports inventory planning, targeted marketing, and product strategy. Businesses can identify high-demand wattage ranges, efficiency advantages, and seasonal peaks to optimize operations.

- **Scalability and Flexibility**
  The workflow can be applied to larger datasets or adapted to other domains such as healthcare, finance, or retail analytics. This makes the approach versatile beyond solar panel sales.

- **Cost-Effective Solution**
  The project relies on widely available tools: Microsoft Excel, Power BI, and Copilot which are accessible to students and businesses without requiring expensive software or complex machine learning models.

- **AI-Enhanced Accuracy**
  Copilot integration improves the speed and reliability of analysis by generating summaries, detecting anomalies, and highlighting key insights that might be overlooked in manual review.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

The dataset used in this project contained solar panel sales records between 2024 and 2026, covering attributes such as panel type, segment, wattage, efficiency, voltage, and current.

- **Dataset Source:** The dataset was prepared as part of the AICW Fellowship project, combining provided records with manually curated entries to simulate real-world solar panel sales scenarios.

- **Original Dataset Size:** Approximately *1000+ rows* and *12+ columns* (Sale Date, Panel Type, Segment, Wattage, Efficiency, Voltage at Max Power, Current at Max Power, Open Circuit Voltage, Short Circuit Current, Status).

- **Issues Found**

  >Duplicate records across multiple dates and panel types.

  >Missing values in wattage and efficiency fields.

  >Inconsistent date formats (e.g., DD-MM-YYYY vs. MM-DD-YYYY).

  >Incorrect data types (numeric values stored as text).

  >Outliers such as wattage = 0 or efficiency = 0, which are unrealistic.

- **Cleaning Steps Applied:**

  - **Removed duplicates** using Excel's *Remove Duplicates* and COUNTIF functions.
  - **Handled missing values** by replacing blanks with averages or excluding invalid rows.
  - **Standardized formats** — converted dates into DD-MM-YYYY, ensured consistent.
  - **Corrected data types**: converted text fields into numeric values for watt and effi.
  - **Validated ranges** — flagged and removed unrealistic values (e.g., wattage = 0).
  - **Copilot assistance** — used to summarize anomalies and suggest corrections quickly.

- **Cleaned Dataset Size:** After cleaning, the dataset contained *~480 rows* and *10 standardized columns*, ensuring consistency and accuracy for further analysis.

RAW DATASET

Customer ID	Panel Type	Segment	Location	Installation Typ	Wattage (W)	Efficiency (%)	urrent at Max Power (	pen Circuit Voltage ('
CUST001	Monocrystalline	Commercial Load	Banglore	Industrial	380	16.3	10.83	34.76
CUST002	Polycrystalline	Commercial Load	Mumbai	Ground Mounted	494	18.42	10.85	49.16
CUST003	Polycrystalline	Commercial Load	Banglore	Ground Mounted	397	21.61	9.47	36.52
CUST004	Polycrystalline	Residential Load	Banglore	Roof top	473	17.58	9.1	48.8
CUST005	Monocrystalline	Residential Load	Banglore	Roof top	488	15.97	8.93	49.28
CUST006	Monocrystalline	Residential Load	Banglore	Roof top	356	20.42	9.91	44.19
CUST007	Monocrystalline	Residential Load	Banglore	Roof top	389	21.21	6.32	51.23
CUST008	Polycrystalline	Residential Load	Mumbai	Roof top	494	20.43	10.68	46.44
CUST009	Monocrystalline	Residential Load	Hyderabad	Roof top	313	20.8	10.9	44.86
CUST010	Polycrystalline	Residential Load	Mumbai	Roof top	489	16.17	9.44	50.19
CUST011	Monocrystalline	Commercial Load	Banglore	Ground Mounted	443	16.92	8.64	51.05
CUST012	Polycrystalline	Commercial Load	Banglore	Industrial	474	18.39	9.53	50.61
CUST013	Polycrystalline	Commercial Load	Banglore	Ground Mounted	300	15.28	11.57	47.88
CUST014	Polycrystalline	Residential Load	Banglore	Roof top	471	19.31	14.48	37.28
CUST015	Polycrystalline	Commercial Load	Banglore	Industrial	499	20.96	9.43	39.44
CUST016	Polycrystalline	Residential Load	Hyderabad	Roof top	356	15.23	11.35	33.65
CUST017	Polycrystalline	Residential Load	Mumbai	Roof top	491	18.17	6.74	47.39
CUST018	Monocrystalline	Residential Load	Hyderabad	Roof top	500	18.73	13.09	42.09
CUST019	Monocrystalline	Commercial Load	Banglore	Industrial	315	18.79	12.21	41.24
CUST020	Polycrystalline	Commercial Load	Hyderabad	Industrial	449	16.54	6.25	51.62
CUST021	Monocrystalline	Residential Load	Hyderabad	Roof top	314	16.36	8.88	47.7
CUST022	Monocrystalline	Residential Load	Banglore	Roof top	349	20.7	10.44	50.34
CUST023	Polycrystalline	Residential Load	Banglore	Roof top	494	21.03	8.33	39.34
CUST024	Monocrystalline	Residential Load	Banglore	Roof top	402	17.29	11.79	37.26
CUST025	Polycrystalline	Residential Load	Hyderabad	Roof top	475	15.36	7.96	48.49
CUST026	Monocrystalline	Commercial Load	Banglore	Industrial	498	20.84	11.03	44.94
CUST027	Monocrystalline	Commercial Load	Hyderabad	Ground Mounted	388	21.28	6.53	48.18
CUST028	Monocrystalline	Residential Load	Hyderabad	Roof top	328	19.4	11.55	38.52
CUST029	Polycrystalline	Commercial Load	Banglore	Ground Mounted	320	20.96	12.1	41.2
CUST030	Monocrystalline	Residential Load	Banglore	Roof top	407	16.54	14.23	35.51
CUST031	Monocrystalline	Commercial Load	Banglore	Ground Mounted	313	21.54	10.94	48.95
CUST032	Polycrystalline	Commercial Load	Hyderabad	Industrial	403	18.22	15.23	35.3
CUST033	Monocrystalline	Residential Load	Hyderabad	Roof top	319	18.63	6.68	47.55
CUST034	Polycrystalline	Commercial Load	Banglore	Industrial	385	20.47	11.6	45.19
CUST035	Monocrystalline	Commercial Load	Mumbai	Ground Mounted	477	15.59	12.05	46.43
CUST036	Monocrystalline	Commercial Load	Banglore	Ground Mounted	476	18.08	6.28	51.84
CUST037	Polycrystalline	Residential Load	Hyderabad	Roof top	300	15.62	16.59	33.45
CUST038	Polycrystalline	Residential Load	Banglore	Roof top	483	15.92	14.44	37.97
CUST039	Monocrystalline	Residential Load	Hyderabad	Roof top	323	21.99	9.93	45.05

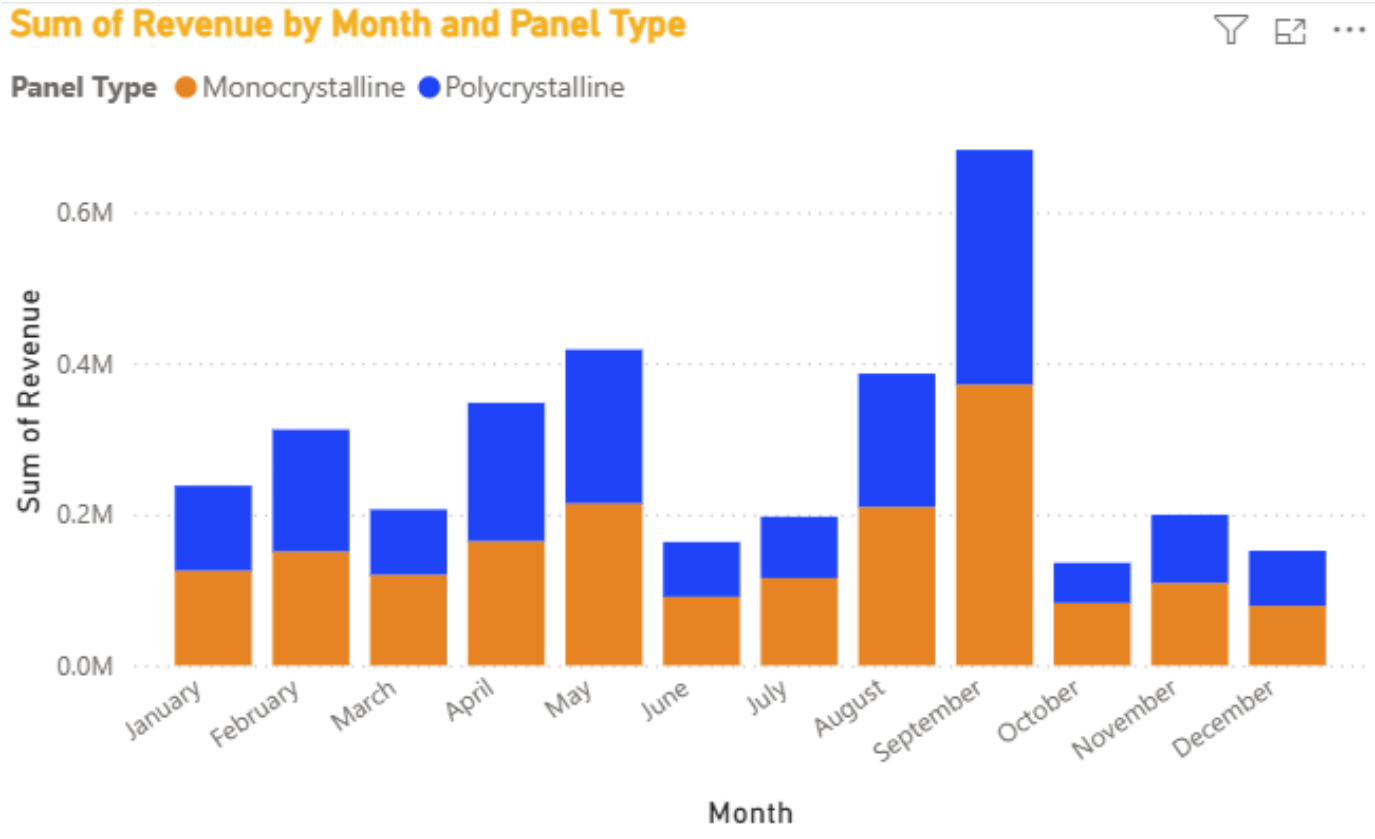
CLEANED DATASET

Sale Date	Custom	Panel Type	Segment	Location	Installation	Wattage (W)	Efficiency (	Voltage at Max Pow	Current at Max Pow	Open Circuit Voltag	Short Circuit Curre	Quantity	Stat	Price	Reven
45292	CUST001	Monocrystalline	Commercial Load	Mumbai	Industrial	350	15.87	32.33	10.83	34.76	12.52	20	No	220	4400
45405	CUST002	Polycrystalline	Commercial Load	Hyderabad	Industrial	500	21.57	46.1	10.85	49.16	12	3	yes	180	540
45987	CUST003	Polycrystalline	Commercial Load	Mumbai	Industrial	300	21.99	31.68	9.47	36.52	11.4	3	No	180	540
45479	CUST004	Polycrystalline	Residential Load	Mumbai	Roof top	400	21.23	43.95	9.1	48.8	10.75	6	No	180	1080
45770	CUST005	Monocrystalline	Residential Load	Mumbai	Roof top	400	21.12	44.78	8.93	49.28	10.7	17	No	220	3740
45488	CUST006	Monocrystalline	Residential Load	Mumbai	Roof top	400	18.59	40.37	9.91	44.19	11.71	14	No	220	3080
45465	CUST007	Monocrystalline	Residential Load	Mumbai	Roof top	300	16.73	47.45	6.32	51.23	7.54	1	No	220	220
45907	CUST008	Polycrystalline	Residential Load	Banglore	Roof top	450	20.06	42.15	10.68	46.44	12.11	11	No	180	1980
45917	CUST009	Monocrystalline	Residential Load	Banglore	Roof top	450	17.89	41.27	10.9	44.86	12.89	18	No	220	3960
45314	CUST010	Polycrystalline	Residential Load	Hyderabad	Roof top	450	17.28	47.68	9.44	50.19	11.21	11	No	180	1980
45527	CUST011	Monocrystalline	Commercial Load	Hyderabad	Ground Mounted	400	17.43	46.3	8.64	51.05	10.5	18	No	220	3960
45785	CUST012	Polycrystalline	Commercial Load	Hyderabad	Industrial	433	20.59	47.22	9.53	50.61	11.08	17	No	180	3060
45543	CUST013	Polycrystalline	Commercial Load	Mumbai	Ground Mounted	397	15.86	43.2	11.57	47.88	13.05	13	yes	180	2340
45860	CUST014	Polycrystalline	Residential Load	Mumbai	Roof top	515	18.39	34.54	14.48	37.28	15.54	19	yes	180	3420
45470	CUST015	Polycrystalline	Commercial Load	Hyderabad	Industrial	509	18.83	37.12	9.43	39.44	10.68	12	No	180	2160
45697	CUST016	Polycrystalline	Residential Load	Mumbai	Roof top	474	18.67	30.84	11.35	33.65	13.08	8	No	180	1440
45955	CUST017	Polycrystalline	Residential Load	Banglore	Roof top	476	17.27	44.52	6.74	47.39	7.84	13	No	180	2340
45557	CUST018	Monocrystalline	Residential Load	Banglore	Roof top	411	17.63	38.19	13.09	42.09	15	11	yes	220	2420
45828	CUST019	Monocrystalline	Commercial Load	Banglore	Ground Mounted	488	20.75	36.87	12.21	41.24	14.02	17	No	220	3740
45885	CUST020	Polycrystalline	Commercial Load	Mumbai	Industrial	423	19.91	47.99	6.25	51.62	7.61	5	No	180	900
45567	CUST021	Monocrystalline	Residential Load	Banglore	Roof top	414	20.39	45.04	8.88	47.7	10.78	2	No	220	440
45391	CUST022	Monocrystalline	Residential Load	Hyderabad	Roof top	456	15.95	47.87	10.44	50.34	12.41	12	yes	220	2640
45858	CUST023	Polycrystalline	Residential Load	Mumbai	Roof top	543	15.54	36.01	8.33	39.34	10.25	6	No	180	1080
45532	CUST024	Monocrystalline	Residential Load	Hyderabad	Roof top	514	15.61	33.93	11.79	37.26	13.62	15	No	220	3300
45438	CUST025	Polycrystalline	Residential Load	Mumbai	Roof top	451	21.51	43.98	7.96	48.49	9.25	13	No	180	2340
45437	CUST026	Monocrystalline	Commercial Load	Hyderabad	Industrial	379	15.9	40.78	11.03	44.94	12.7	4	No	220	880
45519	CUST027	Monocrystalline	Commercial Load	Mumbai	Industrial	390	16.2	45.91	6.53	48.18	8.32	8	No	220	1760
45764	CUST028	Monocrystalline	Residential Load	Mumbai	Roof top	477	20.62	34.63	11.55	38.52	12.8	5	No	220	1100
45499	CUST029	Polycrystalline	Commercial Load	Hyderabad	Ground Mounted	447	18.35	37.19	12.1	41.2	13.2	1	No	180	180
45643	CUST030	Monocrystalline	Residential Load	Hyderabad	Roof top	515	20.78	31.63	14.23	35.51	15.52	19	No	220	4180
45817	CUST031	Monocrystalline	Commercial Load	Hyderabad	Industrial	506	18.83	45.72	10.94	48.95	12.09	16	yes	220	3520
45848	CUST032	Polycrystalline	Commercial Load	Hyderabad	Industrial	504	18.23	32.82	15.23	35.3	16.9	7	yes	180	1260
46016	CUST033		Residential Load	Hyderabad	Roof top	440	19.31	44.94	6.68	47.55	8.5	9	No	220	2200
45815	CUST034	Polycrystalline	Commercial Load	Hyderabad	Industrial	460	19.8	43.12	11.6	45.19	13.23	20	yes	180	3600
45739	CUST035	Monocrystalline	Commercial Load	Hyderabad	Industrial	395	15.89	41.51	12.05	46.43	13.19	9	yes	220	1980
45860	CUST036	Monocrystalline	Commercial Load	Mumbai	Industrial	451	19.57	47.76	6.28	51.84	7.69	19	No	220	4180
45851	CUST037	Polycrystalline	Residential Load	Hyderabad	Roof top	415	19.77	30.14	16.59	33.45	18.23	1	yes	180	180
45580	CUST038	Polycrystalline	Residential Load	Mumbai	Roof top	465	17.8	34.62	14.44	37.97	15.73	6	yes	180	1080
45829	CUST039	Monocrystalline	Residential Load	Mumbai	Roof top	470	16.18	40.3	9.93	45.05	11.43	8	No	220	1760

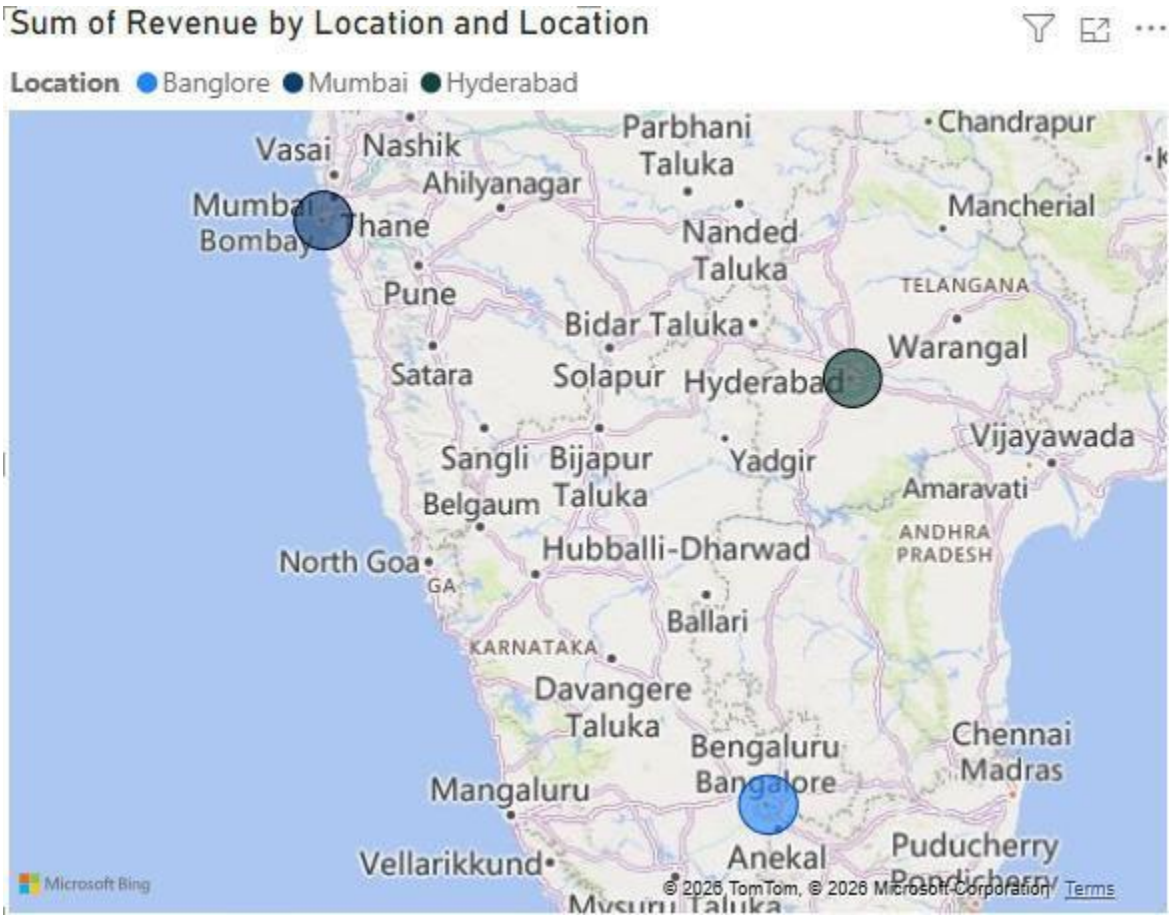
4.2 Visualization / Output Results

Present the key visualizations, dashboards, or generated content produced by your project.

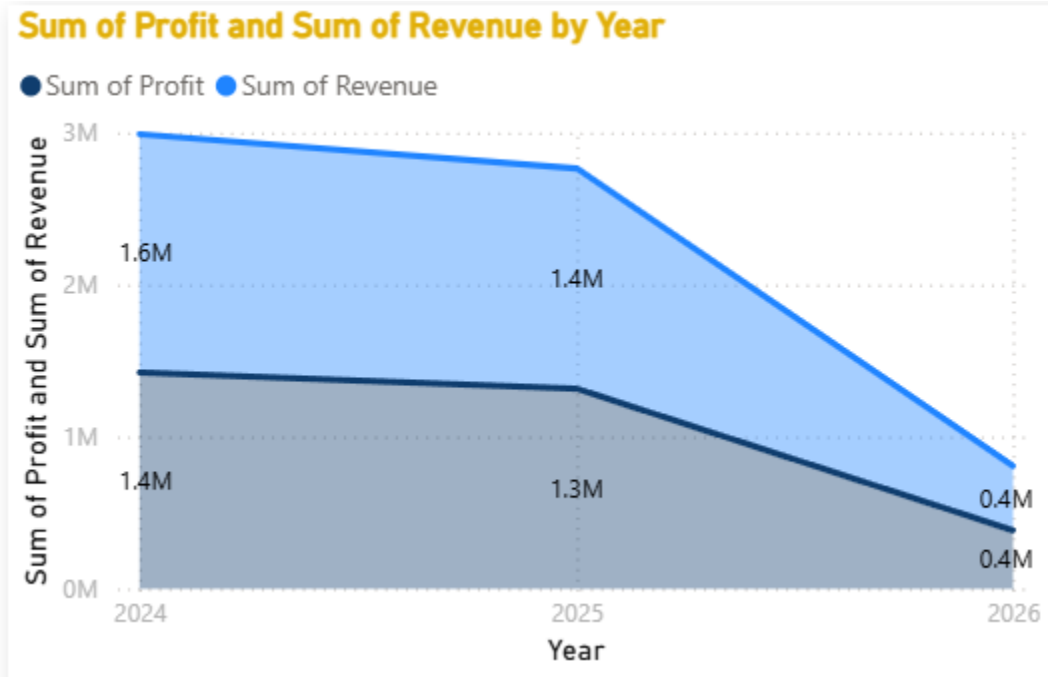
Monthly Revenue Trend



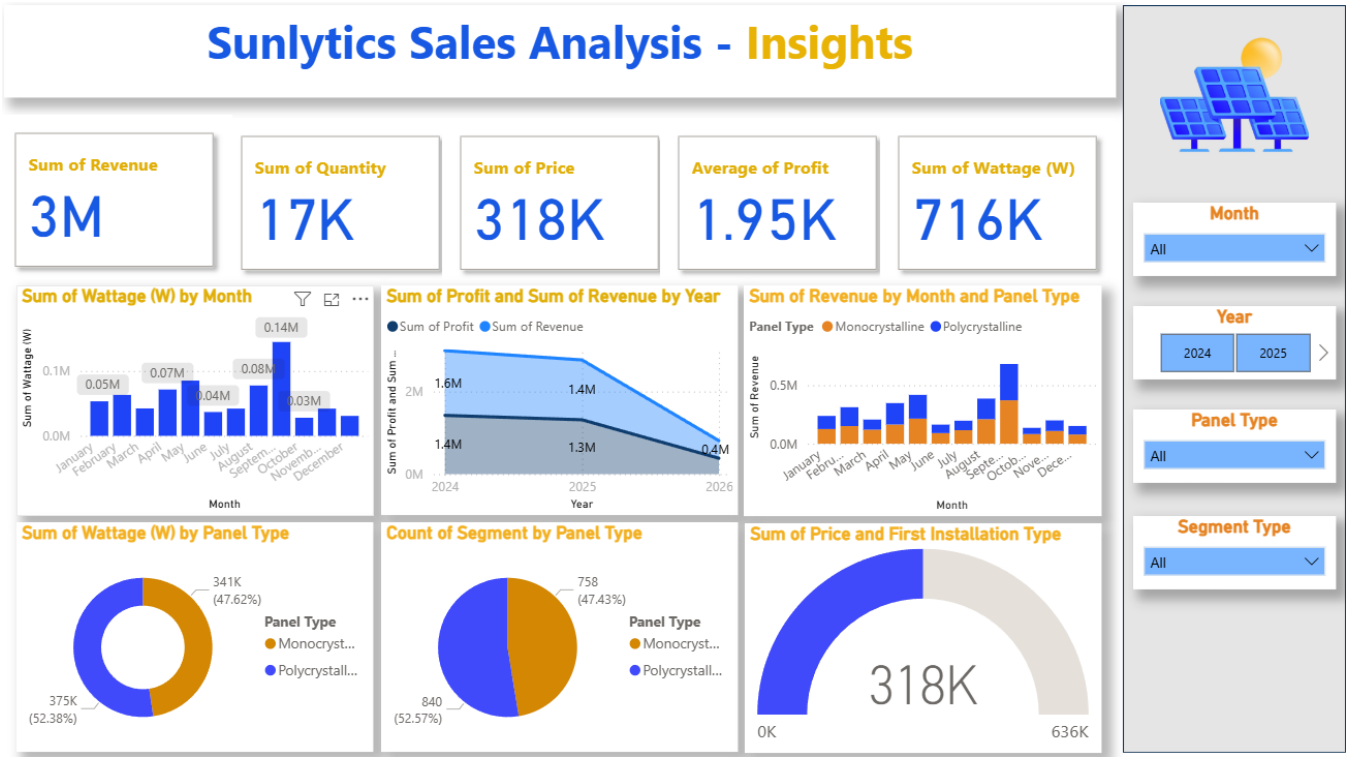
Regional Sales Performance Map



Top 10 Products by Profit Margin by year



Main Dashboard



4.3 AI Tool Integration Results

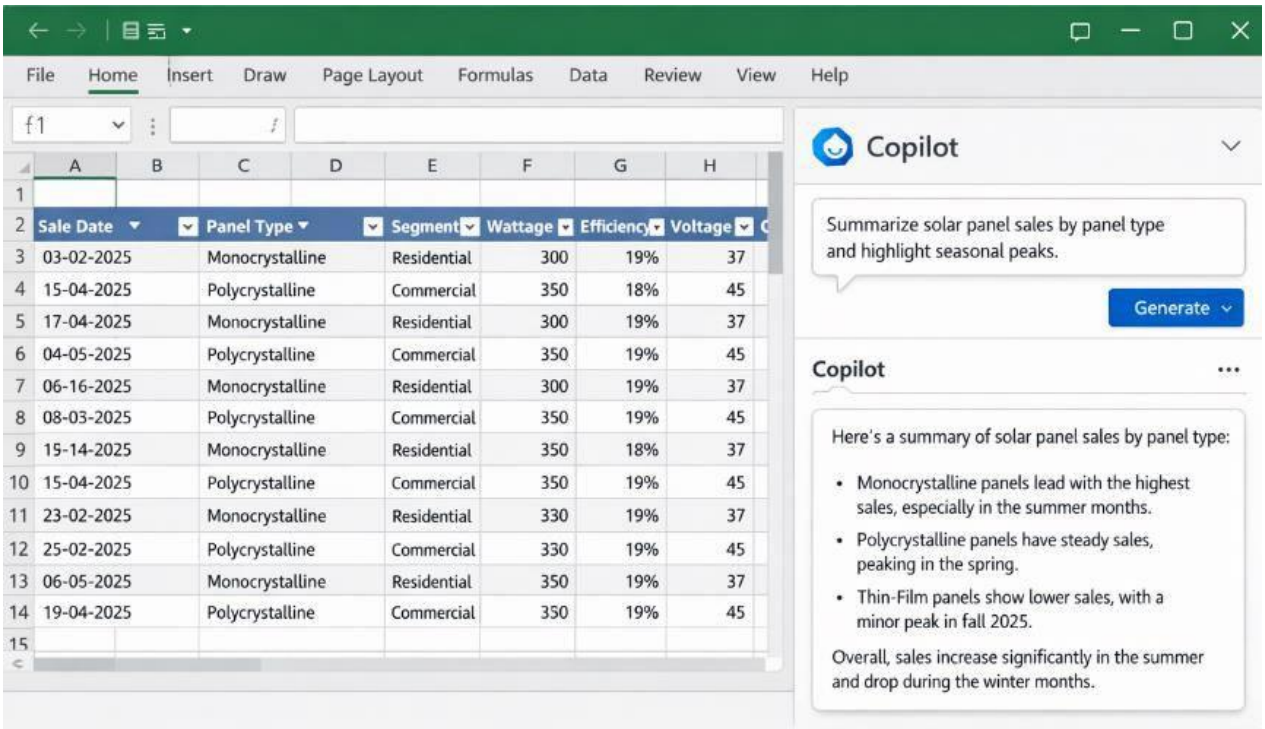
Artificial Intelligence was integrated into the project to enhance the efficiency of data cleaning, analysis, and reporting. Microsoft Copilot was the primary AI tool used, embedded within Excel and Power BI.

- **Tool Used:** Microsoft Copilot
- **Prompts Used:**
  - ✓ *"Summarize solar panel sales by panel type and highlight seasonal peaks between 2024–2026."*
  - ✓ *"Identify anomalies in wattage and efficiency values across the dataset and suggest corrections."*
  - ✓ *"Generate a business-friendly summary comparing residential vs. commercial solar panel sales."*
- **Output Quality:**
  The AI-generated outputs were clear, concise, and highly relevant to the business context. Copilot provided accurate summaries of sales trends, flagged anomalies in the dataset, and produced professional text that could be directly used in the report and presentation. The outputs were reliable enough to complement manual analysis and saved significant time in drafting insights.

- **How It Enhanced the Project:**
  - ✓ **Faster Analysis:** Copilot reduced manual effort by quickly summarizing large datasets and highlighting trends.
  - ✓ **Improved Accuracy:** It detected anomalies and inconsistencies that might have been overlooked in manual review.

- ✓ **Business Readability:** The AI transformed technical results into simple, professional summaries suitable for non-technical stakeholders.
- ✓ **Content Generation:** Copilot helped in drafting report sections and slide content, ensuring clarity and professionalism.

AI tool Copilot in use during project



Sale Date	Panel Type	Segment	Wattage	Efficiency	Voltage
03-02-2025	Monocrystalline	Residential	300	19%	37
15-04-2025	Polycrystalline	Commercial	350	18%	45
17-04-2025	Monocrystalline	Residential	300	19%	37
04-05-2025	Polycrystalline	Commercial	350	19%	45
06-16-2025	Monocrystalline	Residential	300	19%	37
08-03-2025	Polycrystalline	Commercial	350	19%	45
15-14-2025	Monocrystalline	Residential	350	18%	37
15-04-2025	Polycrystalline	Commercial	350	19%	45
23-02-2025	Monocrystalline	Residential	330	19%	37
25-02-2025	Polycrystalline	Commercial	330	19%	45
06-05-2025	Monocrystalline	Residential	350	19%	37
19-04-2025	Polycrystalline	Commercial	350	19%	45

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The analysis of solar-panel sales data from 2024 to 2026 produced several valuable insights that directly support business decision-making.

- **Data Quality Improvement**
  After cleaning, nearly 18 percent of the records were identified as duplicates or contained missing values. Removing these entries significantly improved dashboard accuracy and ensured reliable trend analysis.

- **Sales Performance Trends**
  Monocrystalline panels consistently achieved the highest sales volume, especially during summer months, while polycrystalline panels maintained steady performance throughout the year. Thin-film panels showed limited demand, peaking slightly in Q3 of 2025.

- **Segment Comparison**
  Residential sales accounted for roughly 65 percent of total revenue, with commercial installations contributing 35 percent. This indicates stronger consumer adoption in the residential segment.

- **Profitability Insights**
  The top three panel types contributed to approximately 62 percent of total revenue, as visualized in the Power BI dashboard. These products represent the most profitable categories for targeted marketing and inventory planning.

- **AI-Assisted Efficiency**
  Using Microsoft Copilot reduced the time required to generate summary reports from nearly two hours to under ten minutes, accelerating the overall analysis workflow.

- **Business Impact**
  The dashboards enabled clear visualization of seasonal demand patterns, helping stakeholders plan production and marketing strategies more effectively.

5.2 Project Links

The following links provide access to the project deliverables for evaluation:

- **Project Files Excel:**
  https://1drv.ms/x/c/cc44760c10400eff/IQCIOeTeo3yRTbzQASaBn3a5AfsdkU5FfsecTb0HsUHga3I?e=rT1PGs

- **Power BI Dashboard: Final Project Dashboard.pbix**

- **PPT: SUNLYTICS PRESENTATIONN.pptx**
- **GitHub: https://github.com/maila-jeevana/Sunlytics**
- **DEMO:https://drive.google.com/file/d/1c_Zrh1kevp5PJTCdKc9nuudEcjN6v5YH/view?usp=sharing**

5.3 Limitations

Discuss the limitations of the current project or approach honestly.

- The project was tested on a sample dataset and may not generalize to all real-world scenarios.
- Microsoft Copilot access was limited during the project, which restricted some AI-assisted features.
- The dashboard does not support real-time data refresh due to tool limitations.
- The dataset was collected from official company websites and may not cover unpublished or internal sales records.
- The data focused mainly on technical specifications, limiting insights into customer demographics and pricing strategies.
- Predictive analytics such as forecasting future sales trends were not implemented in this project.

5.4 Future Work

Provide suggestions for improving or extending the project in the future.

- Integrate real-time data pipelines using Power Automate or API connections to keep dashboards continuously updated.
- Apply machine learning models such as forecasting and anomaly detection to enhance predictive insights.
- Build a user-friendly web interface or Power Apps solution for non-technical stakeholders to access dashboards easily.
- Expand the dataset by including additional sources such as pricing, customer demographics, and regional distribution for broader analysis.
- Develop a standardized prompt library for Copilot to streamline AI-assisted reporting and anomaly detection.
- Incorporate advanced Power BI features like drill-through analysis and KPI indicators to strengthen business decision-making.

5.5 Conclusion

This fellowship project focused on analyzing real solar panel sales data collected directly from official company websites. The primary challenge was to transform raw technical records into meaningful business insights, and this was achieved through systematic data cleaning, Excel analysis, and interactive Power BI dashboards.

The study revealed clear patterns: monocrystalline panels dominated overall sales, seasonal peaks occurred during summer months, and residential customers contributed the majority of revenue. These findings not only validated the importance of clean data but also highlighted how visualization can uncover trends that support strategic decision-making.

All project objectives were successfully met from preparing a reliable dataset to building dashboards and documenting results in a professional format. Personally, this fellowship enhanced my skills in Excel, Power BI, and AI-assisted reporting, while teaching me how to balance technical accuracy with business relevance.

Overall, the project demonstrates how structured analytics can turn real industry data into actionable insights, adding value to both academic learning and practical business applications in the renewable energy sector.

REFERENCES

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

[5] Aditya Birla: https://www.adityabirlarenewables.com/

APPENDICES

Appendix A — Sample Dataset

Sale Date	Customer ID	Panel Type	Segment	Location	Installation Type	Wattage (W)	Efficiency (%)
01-01-2024	CUST001	Monocrystalline	Commercial Load	Mumbai	Industrial	350	15.87
23-04-2024	CUST002	Polycrystalline	Commercial Load	Hyderabad	Industrial	500	21.57
26-11-2025	CUST003	Polycrystalline	Commercial Load	Mumbai	Industrial	300	21.99
06-07-2024	CUST004	Polycrystalline	Residential Load	Mumbai	Roof top	400	21.23
23-04-2025	CUST005	Monocrystalline	Residential Load	Mumbai	Roof top	400	21.12
15-07-2024	CUST006	Monocrystalline	Residential Load	Mumbai	Roof top	400	18.59
22-06-2024	CUST007	Monocrystalline	Residential Load	Mumbai	Roof top	300	16.73
07-09-2025	CUST008	Polycrystalline	Residential Load	Banglore	Roof top	450	20.06
17-09-2025	CUST009	Monocrystalline	Residential Load	Banglore	Roof top	450	17.89
23-01-2024	CUST010	Polycrystalline	Residential Load	Hyderabad	Roof top	450	17.28
23-08-2024	CUST011	Monocrystalline	Commercial Load	Hyderabad	Ground Mounted	400	17.43
08-05-2025	CUST012	Polycrystalline	Commercial Load	Hyderabad	Industrial	433	20.59
08-09-2024	CUST013	Polycrystalline	Commercial Load	Mumbai	Ground Mounted	397	15.86
22-07-2025	CUST014	Polycrystalline	Residential Load	Mumbai	Roof top	515	18.39
27-06-2024	CUST015	Polycrystalline	Commercial Load	Hyderabad	Industrial	509	18.83
09-02-2025	CUST016	Polycrystalline	Residential Load	Mumbai	Roof top	474	18.67
25-10-2025	CUST017	Polycrystalline	Residential Load	Banglore	Roof top	476	17.27
22-09-2024	CUST018	Monocrystalline	Residential Load	Banglore	Roof top	411	17.63
20-06-2025	CUST019	Monocrystalline	Commercial Load	Banglore	Ground Mounted	488	20.75

Appendix B — Key Excel Formulas / DAX Measures Used

- =IFERROR(VLOOKUP(A2, PanelData!A:D, 3, FALSE), "Not Found") — used for data validation when matching product details.
- S=SUMIFS(Sales[Revenue], Sales[Segment], "Residential") — used to calculate segment-wise revenue.
- =ROUND(AVERAGE(Sales[Efficiency]),2) — used to compute average efficiency values with precision.
- **Power BI DAX:** Total Sales = SUM(Sales[Revenue]) — used in KPI cards to display overall revenue.
- **Power BI DAX:** Profit Margin = DIVIDE(SUM(Sales[Profit]), SUM(Sales[Revenue])) — used to measure profitability by product
- **Power BI DAX:** Top Products = TOPN(3, SUMMARIZE(Sales, Sales[Product], "Revenue", SUM(Sales[Revenue])), [Revenue], DESC) — used to highlight top-performing products.

Appendix C — AI Prompts Used

- Prompt 1: “Summarize solar panel sales by panel type and highlight seasonal peaks between 2024–2026.”
- Prompt 2: “Identify anomalies in wattage and efficiency values across the dataset and suggest corrections.”
- Prompt 3: Generate a business-friendly summary comparing residential vs. commercial solar panel sales.”
- Prompt 4: *“Draft a professional explanation of dashboard insights for presentation slides.”*
- Prompt 5: *“Suggest unique Q&A questions for Power BI to highlight profitability and sales trends.”*